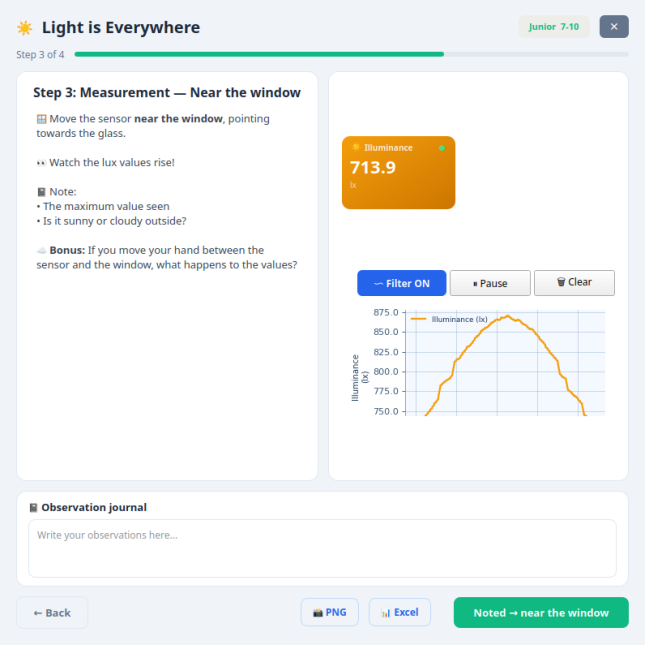
Light is Everywhere

Discovering that “dark” and “bright” are numbers — and that the gap between them is far wider than the eye believes

Level: Junior — ages 7–10 (Romanian school years II–IV) • **Time in class:** 15 min, one full lesson, or 3 short lessons in a week • **Sensors:** The light sensor (illuminance, in lux)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Junior — ages 7–10 (Romanian school years II–IV). How far you can go is decided by the size of the numbers: year II works within 0–1000, so the *under the desk / middle of the room* comparison fits entirely, but the window value does not. Numbers up to 10 000 belong to year III, and those up to a million to year IV.

Assumed knowledge

Comparing two numbers and subtracting. **Decimals are not required** — no number in this activity has a decimal point. The question “how many times more?” needs division and is asked **only from year III**; in year II it stays “much more / a little more”.

Time in class

Short: 15 min. Full: one 50-minute lesson. Project: three short 20-minute lessons in one week (about an hour).

Teacher preparation

10 min — stick labels on the measuring spots, **check the blinds** (if they are half closed the window reading collapses and the whole comparison loses its punch), write the app’s four reference values on the board, and run a one-minute trial.

Sensor settling time

None. This is the only Junior activity with no waiting time, and the biggest single difference from *The Magic Thermometer*: the light sensor responds instantly. Still hold the box still for **5–10 seconds** so the chart line settles, and do not confuse that with a slow sensor coming to equilibrium.

Sensors and boxes	The light sensor in the box, which reports illuminance in lux. One box is enough for the whole class if you work in a semicircle with the screen projected; ideally, one box per 4–6 children.

Three routes

Short	15 min: written prediction, the four reference values on screen, two readings under the desk, two beside the window, the "How sure are we?" step, comparison and conclusion. **Exactly two places.**
Full	The 50-minute lesson: the short route, unhurried, plus the light map of the room with four more places, and the hand test, with either the cloth layers or the coloured papers.
Project	Three short lessons: the two measurements (lesson 1), the light map of the room (lesson 2), the same places at another time of day or in different weather (lesson 3).

1. Where this fits in the curriculum

Children have been saying "it's dark" and "it's light" since they were three. What they have not met is the idea that dark and light are *quantities*, that they can be measured with an instrument, and that between the most shaded corner of the room and the window the difference is not twice but tens of times. Here the sensor does something no discussion can do: it settles an argument with evidence. The activity fits:

- **Mathematics and exploring the environment** (years I–II) — objects and phenomena in the immediate environment, sources of light, shadow, and measuring with an instrument. At this level the numbers stay **below** 1000, so the comparison is made between indoor places; the window value is read aloud by the teacher and discussed as "an enormous amount", without being written down.
- **Natural sciences** (years III–IV) — light and shadow, the straight-line travel of light, the objects that block it, why day is brighter than night, and why plants need light. This is also where the idea of a *fair comparison* settles naturally: the same sensor, pointed the same way, in a different place.
- **Mathematics** (a subject in its own right from year III) — and this is the activity's most unexpected contribution. The numbers the sensor gives are *real*, not invented for an exercise: 47 lux under the desk and 8450 beside the glass. Reading four-digit numbers and putting them in order is exactly the year III syllabus (working within 0–10 000), and the values in direct sun, in the tens of thousands, fall within the year IV range. Subtraction gives the difference; division, from year III, answers the question that matters: *how many times more?*
- **Geography** (year IV only, where the subject appears in the framework curriculum) — weather and cloud, why buildings have windows on particular sides, how the light in the room changes through the day and through the seasons.
- **Language** — describing what is observed in full sentences, and the vocabulary of comparison: *brighter, dimmer, much more, a little less, about the same.* Every place measured gets a word, not just a number.
- **Art and practical skills** — shadow, contrast, and the very practical question of where you sit when you want to draw. This is an *application*, not a syllabus entry: the drawing organises the data.

The core skill is **comparing two measurements**. Children of this age do not need slopes or rates; they need to internalise that a measurement is a number, that numbers can be put side by side, and that the difference between them means something real about the world.

2. Time plan

The in-app activity has only a few steps and fits comfortably into a quarter of an hour — genuinely, this time, because there is no settling time at all. So the short route below is **fifteen** minutes, not twenty as with the thermometer. The risk here is precisely the opposite: because the sensor responds instantly, a class can measure ten places in ten minutes and go home having compared nothing with anything.

Route	Time	What happens
Short	**15 min**	Written predictions (3), the reference values on screen (2), the measurement under the desk (3), the measurement beside the window (4), "How sure are we?" (1), comparison and conclusion (2). These are exactly the steps in the app and **exactly two places measured**, each with two readings.
Full	**50 min**	Written predictions, on the sheet and in the app (7), the reference values and reading large numbers (5), under the desk (5), beside the window (6), the light map of the room with four more labelled places (14), the hand test and one of the material tests (5), "How sure are we?", comparison, drawing and conclusion (8).
Project, lesson 2	20 min	**The light map**: five or six named places — the corridor, the inside of the cupboard, the book corner, the board, the doorway, the sill — with the values marked on a plan of the room drawn on the board. Every group measures the same points, marked with labels.
Project, lesson 3	20 min	**The same three places, another moment**: at a different time of day, or on a cloudy day compared with a sunny one. Here children discover the hardest and most valuable thing — light is not a fixed property of a place.

The project route starts from the short lesson and adds lessons 2 and 3, so it takes **three short 20-minute lessons in the same week** — about an hour of actual time. Children are set no homework and no preparation.

The gift of this activity: the sensor answers instantly

The light sensor has no settling time — and that is its great gift with young children. Do not merely tell them it responds instantly: **show them**. Let a child pass a hand over the sensor and watch the line on the chart dive and recover together. Ten seconds of that does more for their grasp of *the instrument is measuring right now* than any explanation.
But that is exactly why the number of places must be capped. Because nobody is waiting for anything, the temptation is to measure everything. A class that takes ten places in ten minutes is left with ten numbers and no comparison. The short route has **exactly two** places and the full lesson six — all marked in advance with labels, so that every group measures the same points. Allow about a minute per place, no more, or attention will drift.

3. Materials

Item	Per group	Notes
MagicBox with the light sensor	1	The sensor's window must face outwards, not covered by the lid or by anybody's fingers.
Laptop with the app	1	USB cable or WiFi. On a cable, the cable length decides how far from the laptop you can measure.
Long USB cable or extension lead	1	Only if you want to reach the window and the corridor from one laptop position.
Small LED torch	1	For the distance test and for the shadow demonstration.
Sheets of coloured paper	3–4	Held over the sensor: children see that different colours let different amounts of light through.
A sheet of white paper and a piece of dark card	1 each	Placed *under* the sensor, they show reflected light. A dramatic and very cheap comparison.
Small cardboard box or a book	1	To make a "cave" over the sensor — the smallest value of the day.
Thin cloth or a paper towel	1	One layer, two layers, three layers: the simplest countable way to reduce the light.
Sticky labels	4–6	To mark and name the measuring spots, so that every group measures the same points.
Clipboard, coloured pencils	1 / child	The sheet is filled in standing up, and the drawing needs yellow and blue.

Preparation the day before: decide and physically mark the measuring spots with labels, so that groups measure the *same* points and the numbers can be compared. **Check the blinds** — if they are half closed the window reading collapses and the whole comparison loses its punch. Charge the laptops, connect one box and confirm a lux value appears. Write the four reference values from the app's introduction step on the board or on a poster — night, lit room, cloudy outdoors, direct sun — because children will keep looking back at them. And note the value in the middle of the room: during the lesson you will know at once whether a group's number is plausible.

4. Scientific background for the teacher

What the light sensor actually measures

The sensor measures **illuminance**: the amount of luminous flux arriving on each square metre of its little window. The unit is the **lux**, and one lux is one lumen per square metre:

$$1\ \text{lx} = 1\ \text{lm/m}^2.$$

The word "luminous" matters. This is not raw radiant power in watts: the lumen is weighted by the sensitivity of the human eye, which peaks in the green at about 555 nm and falls away towards red and blue. The sensor carries an optical filter that imitates that curve, so its readings track what a person would call brightness rather than what a physicist would call irradiance. Infrared from a radiator or from sunlight is therefore largely ignored — exactly what you want in a classroom.

Two consequences are worth holding in mind while teaching. First, a red torch and a green torch of the same electrical power will *not* give the same lux. Second — and this causes more confusion in class than anything else — the sensor measures the light *arriving at the sensor*, not the light in the room in general. Pointed at the ceiling and then at the floor, from the same spot, it gives two different numbers. Both are correct.

4.1 Why the numbers are so large and so unevenly spaced

The values children will meet span five orders of magnitude:

Situation	Illuminance	Note
Moonless night, outdoors	below 1 lx	The app's first reference point.
Corridor, inside a cupboard	20–100 lx	Enough to walk, not to read.
Under the desk	30–150 lx	The activity's first measurement.
Well-lit classroom	300–500 lx	The recommended level for school work.
Beside the window, overcast	1000–10 000 lx	The activity's second measurement.
Direct sunlight	50 000–100 000 lx	About two hundred times the classroom. The box measures it only with the new firmware (see the technical note).

The reason children are not astonished by this range in daily life is that human vision is approximately **logarithmic**: perceived brightness grows roughly with the logarithm of illuminance, so a hundredfold physical increase feels like a modest change. The pupil adapts as well — in about a second for the fast component, and over several minutes for full dark adaptation.

This is the single most important thing for the teacher to understand, because it explains the moment the activity catches fire: *the window does not **look** twenty times brighter than the desk, but the sensor says it is.* The sensor is not wrong and the children's eyes are not wrong — they are measuring different things, one physical and one perceptual. Said aloud in front of the class, that sentence is the best lesson in method the hour contains.

4.2 Why it is darker under the desk

Light from the window and from the ceiling lamps travels in straight lines. The desktop is opaque, so it blocks every straight path from those sources to the space beneath. What remains underneath is only light that has bounced — off the floor, the chair, the walls — and each bounce absorbs a good part of it. A matt cream wall reflects roughly 70–80% of the light that hits it; a dark carpet perhaps 10%. That is the whole geometry of shadow at this age level, and it can be shown in three seconds with a torch and a hand.

4.3 The inverse square law (for the teacher's own questions)

If a small source radiates equally in all directions, its light at distance d is spread over a sphere of area $4\pi d^2$, so

$$E = \frac{I}{d^2},$$

where E is illuminance and I is luminous intensity in candela. Double the distance and the illuminance falls to a quarter. Children of 7–10 should not be given the formula, but they can absolutely discover the effect: the torch at one ruler-length and then at two ruler-lengths from the

sensor, with both numbers read. In practice expect a fall to somewhere between a quarter and a half, because the room's own light is added to the torch's and because torch beams are directional rather than spherical. Do not present the mismatch as an error; present it as "the room's light is in there too".

4.4 Daylight, cloud and the window

Light outside ranges from around 1000 lux under heavy cloud to above 100 000 in summer sun. Ordinary glass transmits most visible light, but the reading beside a window depends far more on **geometry** than on the glass: whether the sensor sees a patch of open sky, whether the sun is on that side of the building, and whether a blind, a tree or a neighbouring wall is in the way. A north-facing room may never exceed a few thousand lux; a south-facing one in direct sun will exceed every expectation. Both give a good comparison against the desk, and that is all the activity needs.

5. Lesson flow

The in-app activity has six steps: an introduction, a written prediction typed on the screen, the measurement under the desk, the one beside the window, the **How sure are we?** step, and a comparison conclusion that quotes the children's prediction next to the numbers they measured. Below is what the teacher does around the steps. The times are for the 15-minute short route; in the full 50-minute lesson each step is allowed to breathe and the light map is added.

1. **Elicit written predictions first (3 min).** Before a single box is switched on, ask: *where is there more light, under the desk or beside the window — and how many times more?* Every child writes the answer on the sheet and writes or draws *why*. Collect two or three guesses aloud and write the range on the board. **"Twice as much" is the commonest answer, and it is wrong by a factor of ten or more** — precisely what makes the measurement land. The app has a prediction step too: one child types the group's prediction into the box on screen and presses **My prediction is written**; the answer goes into the journal and comes back in the conclusion.
2. **The introduction step on screen (2 min).** Read the four reference values together: moonless night below 1 lux, well-lit room 300–500, outdoors on a cloudy day 1000–10 000, direct sun 50 000–100 000. Ask which of those numbers they think their room resembles today. The activity opens from the **Activities** page, with the **Start** button on the activity card, and the step ends with the **Got it!** button.

Technical note — right before the first reading

The sensor. The light sensor in the box; nothing extra is connected. It reports **illuminance** in lux, and the screen shows the **Illuminance** card with the unit `lx` and, underneath, its chart over the last three minutes. At the Junior level a **light meter** appears next to the card: a bar that brightens from left to right, with four landmarks — *night*, *classroom*, *cloudy day*, *sun* — and a marker at the current value. The bar's scale is logarithmic: each landmark is tens of times more light than the one before, which is how the eye feels it, and children who cannot yet read large numbers can say "we are near *classroom*". In the exported files the column is called `LUX`.

The range depends on the box's firmware. With firmware 2.2.5 or newer the box chooses its own sensitivity and measures from fractions of a lux up to about 120 000 lx — everything you will meet, from the inside of a cardboard box to direct summer sun. With older firmware the sensor worked at one fixed sensitivity and **stopped at about** 3800 **lx**: under the desk and in the room everything was right, but beside a bright window or outdoors the number stuck at the same value however strong the sun. If you see that, reflash the box

(the flashing guide in the `firmware` folder) — the sensor is not faulty. When the reading sits stuck at about 3800 lx, the app shows a yellow strip above the chart by itself: *The reading is stuck near 3800 lx… Ask your teacher.* The resolution at the bottom end is far below one lux with both versions, so even the "cave" under the cardboard box gives a number rather than zero — worth using deliberately, because that is exactly what demolishes the idea that dark means "no light". Above about 10 000 lx the absolute values are approximate (roughly ±15–20%): the relative comparisons (window against desk) count, not the exact lux.

How long you wait. Almost nothing. The box sends one light value a second, and after a big jump — from under the desk straight into the full window — it needs a second or two to adjust its own sensitivity. It is the only Junior activity with no settling time. Still hold the box still for **5–10 seconds** before reading, so the chart line settles, and read a round value rather than the last trembling digit.

The buttons you will use. The activity advances with the green button at the bottom right, and **its label changes at every step**, because it repeats what you have just finished: **Got it!**, **My prediction is written**, **Noted → under the desk**, **Noted → near the window**, **Continue** (after the **How sure are we?** step), and at the last step **Finish**. Under the chart, at the two measuring steps, sits the **Capture** button: it keeps the number on the card, and the app asks for **two** readings in each place. If a child moves on without capturing, the green button turns pale and the hint *press Capture when the number stops changing* appears below it — you can still continue, but the conclusion will then show a dash instead of the number. The **Back** button, on the left, goes back one step and keeps what was written in the journal at each step. Because this is a Junior activity, the smoothing, pause and clear buttons **do not appear** above the chart, and values are shown as whole numbers — the screen stays as simple as possible. To show the class the dive a hand produces, repeat the gesture slowly twice: the chart keeps the last few minutes on screen, so the dip stays visible while you talk about it.

Exporting the data. At the measurement steps two buttons appear: **PNG** saves the chart as an image, and **Excel** saves to an `.xlsx` file the step's raw readings (unfiltered, with an `elapsed_s` column — seconds since the step began) together with the journal. The children's journal is saved on its own anyway, in the `logs` folder, and at **Finish** the app also writes one Excel file for the whole exercise. **There is no CSV button inside the activity.** The CSV file is taken from the app's **Data & Export** page, with the **Excel** button on the box's row, which opens the **Export data** dialog, where CSV can be chosen; when the app runs in Romanian the CSV uses ";" as the separator and the decimal comma, so that it opens correctly in a Romanian Excel. Save the file following the pattern `class_date_activity` — for example `3B_2026-09-14_lumina_pretutindeni`. Note on the sheet the place measured (the name on its label), **which way the sensor was pointing**, the time and the weather outside, and check that the file really was saved before closing the app.

3. **The measurement under the desk (3 min).** The sensor goes under a desk, window upwards, and stays still for about ten seconds while the line settles. No hands over it. One child presses **Capture**, waits a few seconds and presses it again: two readings of the same place. Every child writes the number and one word describing the light. Ask the app's own question aloud: *why is it darker under the desk than above it?* Accept "the desktop blocks the light" and push once more: *blocks it from where?*

4. **The measurement beside the window (4 min).** The sensor is carried to the window, pointed at the glass. Children watch the value climb as the box crosses the room — that journey is worth narrating aloud. The box then stays still for ten seconds beside the glass and **Capture** is pressed twice again. Record the value and whether it is sunny or cloudy outside. Then do the bonus the app suggests: pass a hand between sensor and window and watch the chart fall and jump back. This is the moment to say: *you have just made a shadow, and we measured it.*

5. **How sure are we? (1 min).** This app step puts the two readings from each place side by side with how far apart they are. At Junior the message is simple: *if the two numbers are close, the reading is good; if they are far apart, something moved.* The app then compares the jump from the desk to the window with that gap and says whether the jump is "real, not luck". Read the sentence aloud: it is a first lesson in uncertainty, without a single hard word. (At the Explorer and Scientist levels the same step asks for two boxes side by side, or for the spread and a control measurement; the activity, however, is designed for Junior.)

6. **Compare and conclude (2 min).** Put the two numbers side by side. Do the subtraction as a class and, if the group is ready, the division too: *the window is about _____ times brighter.* Close with the app's conclusion step, whose green button reads **Finish**, and with its three links to the world outside — plants need light to make their food, too little light indoors is bad for people, and solar panels turn this same light into electricity. Finish by asking every child to complete the conclusion sentence on the sheet.

If you have twenty minutes more

Four short tests of about three minutes each, in rising order of surprise. **The cloth layers**: one, two, three — the light falls by roughly the same ratio each time, not by the same amount, and older children can see the pattern. **White paper and dark card**, placed under the sensor in turn: the difference between them is reflected light, and it is large. **The coloured papers** held over it in turn: a different amount passes through each colour. **The torch near, then twice as far away**: the fall is far greater than anybody expected. Every one of these produces a number the child can compare with a number they already have — which is the entire skill being taught.

The investigation, in one table

The question. Is the light the same everywhere in our room?
The hypothesis. Written on the sheet **before** any measuring: which place will be brighter and, in years III and IV, *how many times* brighter, plus a reason in a few words. It is not changed afterwards.
What I change. One thing only: the **place** the box sits in.
What I measure. Illuminance, in lux, as shown on the card on screen.
What I keep the same. The same box, **the same direction the sensor's window points** — this is the thing most often forgotten and it ruins more comparisons than anything else — the same person reading the screen, no hand and no head over the sensor, and the same lights on or off at every place.
How many repeats. Two readings at each place, a few seconds apart. If they are close, the value is good; if they differ a lot, somebody moved the box or a shadow passed over it.
What makes a reading valid — and what makes it an artefact. Valid is a reading taken with the sensor resting on a surface, pointing in the agreed direction, with nothing above it. An artefact is anything else: a finger on the window takes the value nearly to zero, a child leaning over casts a shadow without meaning to, and the sensor pointed at the floor instead of the glass gives a number several times smaller in the same place. The class rule: place it, point it, step back, read. **If two groups get very different values for the same labelled place, the cause is almost certainly the sensor's direction, not the sensor** — agree the direction as a class and measure again.

6. Analysis after the activity

6.1 What the data should look like

Place	Typical value	What it means
Under the desk	30–150 lx	Only bounced light reaches here.
Middle of the room, on a table	200–500 lx	Ceiling lights plus daylight; the level classrooms are designed for.
Beside the window, overcast	1000–5000 lx	Ten to twenty times the desk reading.
Beside the window, direct sun	10 000–60 000 lx	Can be a hundred times the desk reading.
Inside a closed cardboard box	below 10 lx	The smallest number of the day — and still not zero.
A hand passing over the sensor	falls to 10–30%	A shadow, measured.
One layer of cloth, then two, then three	each layer cuts about the same fraction	Not the same amount: the pattern is multiplication, not subtraction.

Absolute values depend on the weather, the time of day, which way the room faces and whether the lights are on, and with old firmware anything above about 3800 lx shows as 3800. Do not look for a "correct" value. What must come out reliably is the **order**: cardboard box < under the desk < middle of the room < window. Every group should get that order, and comparing the different numbers groups obtain for the *same* place is itself a very good discussion about why measurements differ.

6.2 From two numbers to "how many times more", step by step

Subtraction is within reach, but here it says very little: "8450 − 47 = 8403 lux" is true and unmemorable. The question that stays with the child is *how many times*, and that needs division, so it is asked from year III upwards. Do it on the board, with rounded numbers, in three steps:

1. **Round, so we can work it out in our heads.** 47 becomes "about 50", and 8450 becomes "about 8000". Say openly why you are allowed to: for the question "how many times", a few per cent makes no difference to the answer.
2. **Divide.** 8000 : 50 = 160. If that division is too hard, go by repeated doubling: 50, 100, 200, 400, 800, 1600… — how many times did we double before we got close? That route works very well in year III.
3. **Say it aloud, in words.** "Beside the window there is a hundred and sixty times more light than under the desk." Then ask the golden question: *did your eyes think it was a hundred and sixty times brighter?* No. That is where it all ties together: the eye compresses, the instrument does not.

In year II, where division is not yet in the syllabus, stop at the wording: *much more*, and the comparison against the reference values on the board ("it looks more like 'outdoors on a cloudy day' than 'a lit room' ") does the same work with no arithmetic at all.

6.3 Questions to ask afterwards

1. Under the desk it was darker, but you could still see. Where did that little bit of light come from, if the desktop was in the way?

2. Your eyes did not believe the window was tens of times brighter, but the sensor said it was. Who is right — your eyes or the sensor?
3. If we measured the same two places tonight after dark, with the lights on, which number would change most and which would hardly change?
4. A plant on the window sill and a plant under the desk: which one grows better, and what do our numbers say about why?
5. If you were choosing where to read at home, where would you put your desk now, and which measurement made you decide?

6.4 Assessment ideas

- **Low stakes:** the child circles "dark", "medium" or "bright" on the sheet for each place and draws the two places, colouring the bright one lighter. An adult or a partner reads the numbers aloud.
- **Standard:** the child writes both numbers correctly, circles the larger, finds the difference by subtraction, and writes one full sentence explaining why the window value is bigger, using the word "blocks" or "shadow".
- **Extension:** the child works out roughly *how many times* brighter the window is, predicts in writing the value for a third place before measuring it, then compares prediction with measurement in writing and explains the mismatch.

Common pitfalls and misconceptions

Fingers on the sensor. By far the commonest failure. A thumb resting over the little window takes the reading nearly to zero, and the child concludes the box is broken. Show everyone the "fingers off" rule in the first minute, and **make the mistake deliberately once** so they recognise the symptom.

"Dark means no light at all." Children will say confidently that under the desk there is "no light". The sensor says otherwise — it reads 50 or 100 lux, not zero. Use the closed cardboard box as the second case: *that* is what almost no light looks like, and even there it is rarely zero.

"The sun is hot, so hot places are bright." At this age light and heat get merged. The radiator is warm and dark; the fridge light is bright and cold. If you also run *The Magic Thermometer*, contrast the two directly.

The sensor's direction. Two children measure "at the window" and get 900 and 6000 lux, because one pointed the sensor at the glass and the other at the floor. **This is not an error to be corrected away but a genuine discovery** about what illuminance means. Agree the direction as a class before measuring — and then show the difference deliberately, because it is worth it.

Reading large numbers wrongly. 8450 read aloud as "eight hundred and forty-five" wrecks the comparison. Spend a moment on place value, with the number on the screen in front of you.

Chasing the last digit. The last digits jitter constantly, and some children will try to record an exact value. Teach them to say "about" and round: "about four hundred".

Too many places. Because nothing has to be waited for, the temptation is to measure everything. Ten places in ten minutes means ten numbers and no comparison. Two places, compared seriously, are worth more.

Safety and classroom management

The sensor is never pointed at the sun, and nobody looks at the sun. This is the one real hazard in the activity. The instruction is to measure *beside* the window; it is not to stare at the sun. State the rule explicitly before anybody goes to the window, and stand between the class and any window where the sun is directly visible.
The torch is never pointed at anybody's eyes. Make a rule of it: torches point down, at the table or at the sensor. Cheap LED torches are bright enough to be genuinely unpleasant at close range.
Cables. A child carrying the box on a cable across the room will trip on it or pull the laptop off the table. Use WiFi where you can, or tape the cable down, or carry the laptop with the box. Keep bags and chairs out of the walking path to the window.
Windows stay closed in this activity, and children do not climb on sills or radiators to reach them. If your sill is at a height that invites climbing, measure from a table pushed close to the window instead.
The box is school equipment carried by children: it is held with two hands and walked, not run, across the room.

Adaptations for pupils with special educational needs

No numbers at all. The child records no value: they circle one of the three pictures on the sheet — dark, medium, bright — for each place, and draw the two places with coloured pencils. An adult or a partner reads the value aloud. The learning objective is that light differs from place to place and that a machine can tell us so, and it is met in full with no digits.
Large numbers are the real barrier here. Unlike temperature, this activity produces four- and five-digit numbers. For the child who cannot handle them, split the problem: they write only the under-the-desk number (two or three digits, well within reach) and circle "much more" for the window. The comparison is made in words rather than figures, and it remains a real comparison.
With a single box, the whole class. Seat the children in a semicircle with the screen projected and carry the box yourself from place to place. At each place name a **screen reader**, who says the number aloud, and a **direction keeper**, who checks which way the window points and has the right to stop a crooked reading. The roles rotate at every place.
Three real roles for children with motor difficulties. Not being able to carry the box does not mean not taking part. The roles rotate at every place:

- **the direction keeper** — decides and checks which way the sensor is looking and gives permission to read; this is the role with the greatest scientific value in the group, because direction is precisely what ruins most comparisons;
- **the screen reader** — says the number aloud, rounded, and announces when the line has settled;
- **the mapper** — draws on the sheet where in the room the measured place is, and colours it lighter or darker.

The box can be carried by a classmate on the child's instruction: "point it more towards the glass" is an experimental decision too, and whoever makes it is running the experiment.
Support with writing. Spoken answers are accepted instead of written ones, and the teacher or a classmate writes them on the sheet. For the prediction, the child can tick one of the printed boxes rather than write a free answer. The sheet comes with its labels, rows and drawing boxes already printed for exactly this reason.
Sensory sensitivity and light. A few children find bright light unpleasant, and a few are photosensitive. Do not send them to the window measurement and do not point the torch in their direction; the screen reader and mapper roles are just as valuable and can be done

from a desk. Announce in advance that the torch will be switched on, so nobody is taken by surprise.

If things do not go to plan

You have one box and thirty children. Semicircle, screen projected, and carry the box yourself. Two places, with a screen reader and a direction keeper named at each. The table is filled in together on the board and copied onto the sheets. Because the sensor answers instantly, this version works particularly well: the box's journey across the room, with the value climbing on screen, is itself the spectacle.

You have fifteen minutes, not fifty. This is the short route and it works, on one condition: **exactly two places** — under the desk and beside the window. Drop the light map, the cloth layers and the torch. The written prediction at the start and the comparison at the end are all the hour needs to have a result.

You have no projector. Seat the class in a semicircle round the laptop at a distance from which the screen can be seen, and name a **screen reader** at every place. It works just as well, and sometimes better: somebody has to say each value out loud.

The day is gloomy and the window gives little. That is not a problem but another lesson. Even in rain, beside the glass there are usually 500–2000 lux, still ten times the value under the desk, and the comparison survives intact. Note the weather on the sheet and, if you can, repeat the measurement on a sunny day: two numbers for the same place on two different days are exactly what lesson three of the project route is.

All the numbers come out very small, wherever you measure. Check the obvious first: the lid, a finger, a label stuck by accident over the sensor's window. If the value stays near zero even in full glass, change the box.

At the window the number stops at about 3800 **and will not climb.** The box has old firmware with a fixed sensitivity. The comparison with the desk still holds — the window still comes out tens of times brighter — but tell the children the true number is larger, and reflash the box with the current firmware (2.2.5 or newer) before the next lesson.

The box will not connect. Do not stop the lesson. Use the "What the data should look like" table in this guide as a printed dataset: children write their predictions first, then receive the values row by row and check whether they guessed the order of the places. The discussion about why the eye does not feel the difference the instrument shows is unchanged — and it is the best part of the activity anyway. Measure again another day, after checking the cable, the port selected in the app and, if you work over WiFi, the network.

7. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are base addresses, not deep links, so they do not rot.

- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for the light and colour simulations; excellent to show on the board right after the children have measured for themselves how much passes through a coloured paper.
- **Institute of Physics** — `https://www.iop.org` — search their primary teaching material on light and shadows; it is written for teachers who are not physics specialists and answers exactly the "but why?" questions this activity provokes.
- **STEM Learning** — `https://www.stem.org.uk` — search for light and shadow for ages 7–11; a large library of lesson plans this measurement slots straight into.
- **Romanian National Meteorological Administration** — `https://www.meteoromania.ro` — search for cloud cover and hours of sunshine; useful for lesson three, when a sunny day is

compared with an overcast one.

- **NASA Climate** — `https://climate.nasa.gov` — for the follow-up conversation about sunlight reaching the Earth and about solar energy, at a level that suits a display or an assembly.
- **Royal Society of Biology** — `https://www.rsb.org.uk` — for the link to plants and light, if you go on to grow seedlings in the bright and dim places the children have just measured.

8. Assessment

The exit ticket, identical to the one on the pupil sheet: *one thing that surprised you today.* Collected in the last two minutes of the lesson, it says more about the class's understanding than any mark.

For a more formal judgement, use the **common three-axis rubric** — *measurement practice*, *data interpretation* and *scientific reasoning.* At Junior level the three axes look like this:

- **Measurement practice** — the child keeps fingers away from the window, points the sensor in the agreed direction, and records what the screen shows rather than what they expected it to show.
- **Data interpretation** — the child says which place was brighter and by how much, and from year III upwards roughly how many times.
- **Scientific reasoning** — the child explains, in their own words, why less light reaches under the desk, and compares their conclusion with the prediction written at the start.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. Pupil sheet

The sheet the children fill in is a separate document: `docs/en/student_sheets/light_is_everywhere.pdf`. It has two pages — the prediction, the reference scale and the drawing box on the first; the measuring and the conclusion on the second — so it is printed double-sided, one sheet per child. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write — and it asks for the prediction **before** any measuring.

Four things to know before printing it. **The table has two reading columns**, because the app asks for two captures in each place, and a line underneath (*How sure are we?*) compares the gap between them with the jump from the desk to the window. **The table has a column for the sensor's direction**, not just for place and value: without it, two groups can record very different numbers for the same place and nobody can say why afterwards. **The "dark / medium / bright" column** is there precisely for the child who does not work with four-digit numbers: they circle it and leave the number column empty. **The working-out row carries "how many times" as well as the subtraction**, but with a printed note that it is completed from year III upwards — in year II it stays empty without the sheet looking unfinished. The table is not repeated here precisely so that the two documents cannot drift apart: there is one printable version, and it is the sheet.